



Republic of the Philippines
Department of Agriculture
BUREAU OF FISHERIES AND AQUATIC RESOURCES
REGIONAL FISHERIES LABORATORY XII
J. Catolico St., Lagao, General Santos City

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Introduction

This procedure deals with matters pertaining to handling, checking, transport, storage, safe operation, maintenance and use of laboratory equipment, media and reagents, glassware, reference materials and software.

1. Responsibilities

1. Laboratory Manager
2. Quality Assurance Manager
3. Technical Manager
4. Administrative Officer
5. Laboratory Analyst

2. General

2.1. Each laboratory unit uses only equipment under its permanent control.

2.2 All equipment used for a particular analysis is capable of achieving the measurement accuracy and measurement uncertainty to provide a valid result. This is accomplished through:

- Implementing calibration schedule for all measuring instruments or equipment requiring calibration (LF 03-03, Calibration Schedule).
- Creating the necessary conditions for achieving declared, or stated uncertainty of measurement
- Carrying out frequent intermediate checks where possible
- Operating and applying the present Quality management system etc.

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3. Receipt of new equipment

3.1. All newly received equipment is checked to ensure that it complies with all pertinent standards and specifications set upon ordering or declared by its manufacturer (accuracy class, uncertainty of measurement, tolerance, etc.).

3.2. Measuring equipment is delivered by the supplier calibrated.

3.3. If it is delivered non-calibrated, it must be calibrated before use, under the responsibility of the laboratory analyst, either by submitting it to a qualified calibration laboratory, or through internal calibration.

4. Use and operation of equipment

4.1. Equipment is used as prescribed by its manufacturer. It is protected from misuse and mishandling and is regularly maintained to ensure proper operation.

4.2. Equipment can be used only by approved and adequately trained personnel, according to specific instructions.

4.3. Use of equipment by non-trained personnel is only possible under the close supervision of the immediate superior.

4.4. Documents and records pertaining to equipment installation, operation, servicing etc. are retained by Technical Manager, and are available to all personnel carrying out measurements.

5. Records

5.1. Each laboratory maintains a list and records of the equipment it uses in the form LF 03-01 List of Laboratory Equipment with the following information:

1. Equipment ID
2. Type of Equipment
3. Model/ Brand
4. Serial Number
5. Working Range
6. Evidence of verification that equipment conforms with specified requirements (i.e. accuracy)

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7. Current Location
8. Calibration Program (Frequency; Responsibility)

5.2. Each laboratory maintains a list and records of the equipment it uses in the form LF 03-02 Equipment History with the following information:

1. Date of withdrawal
2. Type of Service, Reason (Calibration/ Maintenance/ Repair)
3. Date Returned
4. Date of Next Service
5. History record of damages, malfunctions, modifications, repairs, etc.
6. History of calibration records with dates, calibration laboratory, calibration certificate number etc.

5.3. The equipment ID for each equipment in a specific laboratory unit will be assigned with the code:

XXX-YYY

Where:

XXX stands for the first three letters of the laboratory unit: CHE- Chemistry Laboratory, MIC-Microbiology Laboratory and FIS-Fish Health Laboratory, etc.

YYY stands for the number of the equipment

6. Monitoring of Equipment

6.1. Every measuring equipment, or part of equipment, destroyed, subjected to overloading or mishandling that gives questionable results, or has been shown to be defective or outside specified requirements, is immediately labeled as "Out of Service" and isolated until it has been verified to perform correctly.

6.2. The laboratory examines the effect of the defect or deviation from specified requirements. It is treated as non-conformance and reported in LF 10-01, Non-conformity Form.

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6.3. All instrument's removal from the laboratory's premises (i.e. for repair and disposal) becomes part of its history record in form LF 03-02, Equipment History.

6.4. All equipment requiring calibration is used only for the time period during which calibration is valid. In case that non-calibrated equipment was used to carry out measurements, the Quality Assurance Manager applies the appropriate corrective actions as prescribed by SOP-17, Corrective Actions. Furthermore, all necessary measures are taken to ensure that equipment continues to measure within its prescribed uncertainty levels. An indicative list of immediate corrective actions this case, comprise the following:

- Vivid labelling to prevent accidental use
- Examination of the effect of the problem on previous tests
- Communication with all customers who may have been affected, with the purpose of notifying them of the problem

6.5. Equipment maintenance is performed depending on the maintenance task description found in LF 03-06, Equipment Maintenance Plan. Performance of equipment maintenance and filling up of form is as follows:

6.5.1. Determine the maintenance task to be performed on the equipment along with the frequency of task (by the number of days).

6.5.2. The person responsible is the one who does the task. It can be internal (by the Technical Manager or Laboratory Analyst or external (by supplier or service engineer).

6.5.3. The condition of the equipment must be indicated (i.e. running or shut down) when maintenance tasks are performed.

6.5.4. The type of maintenance task is also identified. It can be Preventive Maintenance, Predictive Maintenance, Corrective Maintenance or Situational Maintenance based on meter hours, analysis results or special events.

6.5.5. Estimated time of hours to complete the task is entered.

6.5.6. Any special tools, hazardous waste container or personal protective equipment that is needed to perform the task are identified.

6.5.7. The total annual hours required to perform each specific maintenance task is also indicated.

6.6. Calibration certificates or user's manual are used to determine if equipment conforms to specified requirements.

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6.7. Calibration of equipment is scheduled on a yearly basis, before the date of calibration of the previous year.

6.8 The Technical Manager and analyst conduct intermediate checks or system configuration on equipment prior to usage.

7. Repair of Equipment

7.1. Laboratory equipment not functioning within its proper specifications will be labeled as "For Repair" and included in the Equipment History.

7.2. Supplier for equipment repair will be contacted. Repair of equipment can be within laboratory premises or depending on the conditions of the supplier.

7.3. For equipment extracted in the laboratory premises, supplier must take full responsibility on the handling and return of equipment.

7.4. Upon return, the Technical Manager will test the equipment to check and evaluate if equipment is functioning within its specifications. Supplier must submit a service report.

8. Disposal of Equipment

8.1. Laboratory equipment beyond repair is subject to disposal.

8.2. It is reported in the LF 05-07, Property Return Slip, which is submitted to the General Service Unit of the BFAR XII.

8.3. The General Service Unit will set a pull-out schedule of the equipment for disposal.

9. Use of Hardware (HW) and Software (SW)

9.1. All computers or automated equipment for taking, processing, analyzing, handling, recording, etc. of data are safeguarded by Technical Manager and/or Laboratory Analyst safeguards proper operation of that equipment, through the following measures:

- Use of Antivirus Software
- Use of passwords for accessing computer files or programs, if necessary

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- Backing-up of files
- Checking and Software verification, when applicable
- Restricting and preventing access of unauthorized personnel to particular IT equipment, etc.

10. Receipt of New Reagents, Media and Consumables

10.1. Each laboratory unit submits the list of reagents, media and consumables that conforms to specifications of technical requirements.

10.2. Upon receipt of newly delivered reagents, media and consumables, the Purchase Service Officer and Technical Manager inspects if the supplies meets the specifications stated in the Purchase Request.

10.3. If the delivered reagents, media and consumables do not meet the specifications, it is automatically rejected. It should be replaced as what is stated in the Purchase Request.

11. Inventory and Usage of Reagents, Media and Consumables

11.1. The laboratory analyst fills out the LF 05-01 Inventory Form, , for every accepted and consumed reagents, media and consumables.

11.2. A master list of all reagents, media and consumables is updated, as needed.

11.3. The laboratory analyst fills out the Reagents and Media Logbook, LF W01-01, after every use.

11.4. The laboratory sets the ordering point. If the amount reaches the ordering point, the laboratory analyst or Technical Manager reports to the Purchase Service Officer for the purchase of new supplies.

12. Receipt of New Reference Materials and Cultures

12.1. Each laboratory unit submits the list of reference materials and cultures that conforms to specifications of technical requirements and are traceable to ISO/IEC 17034 and American Type Culture Collection (ATCC).

12.2. Upon receipt of newly delivered reference materials and cultures, the Purchase Service Officer and Technical Manager inspects if the supplies meets the specifications stated in the

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Purchase Request.

12.3. If the delivered reference materials and cultures do not meet the specifications, it is automatically rejected. It should be replaced as what is stated in the Purchase Request.

13. Inventory, Usage and Maintenance of Reference Materials and Cultures

13.1. The Technical Manager assures that the reference materials and cultures are maintained in accordance to the instructions from the manufacturer.

13.2. The laboratory analyst fills out the Inventory Form, LF 05-01, for every accepted and reference materials and cultures.

13.3. A master list of all reference materials and cultures is updated, as needed.

13.4. The Technical Manager and laboratory analyst fills out the Maintenance of Cultures, LF 09-03-01, after every use.

13.5. The laboratory sets the ordering point. If the amount reaches the ordering point, the laboratory analyst or Technical Manager reports to the Purchase Service Officer for the purchase of new supplies.

14. Forms

LF 03-01, List of Laboratory Equipment

LF 03-02, Equipment History

LF 03-03, Calibration Schedule

LF 03-05, Equipment Usage and Maintenance Logbook

LF 03-06, Equipment Maintenance Plan

LF 05-01, Inventory Form

LF 05-08, Property Return Slip

LF 10-01, Non-Conformity Form

SOP-17, Corrective Actions

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15. Document History

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